

Solution Blueprint: Aerospace Repair Engineering Order (REO) Compliance Review Agent

Automated Airworthiness Audit for Commercial Aircraft Fleet Support

Hybrid Cognitive Architecture Combining Deterministic Engines & Multimodal LLMs on Google Cloud

Executive Document Information

Attribute	Specification
Client / Domain:	Commercial Aircraft Manufacturing & Engineering Fleet Support
Solution Category:	Airworthiness Compliance Automation (Repair Engineering Orders – REOs)
Technology Stack:	Google Agent Development Kit (ADK) 2.0, Gemini 3.7 Flash Multimodal, Vertex AI Agent Engine, Google Cloud Storage, Gemini Enterprise
Hosting Region:	Google Cloud <code>europa-west1</code> (Belgium) / Runtime: Vertex AI Reasoning Engine
Document Classification:	Enterprise Solution Blueprint & Validated Reference Architecture
Status:	Active in Production / Validated Reference Architecture
Version:	1.0.0 (Anonymized Enterprise Distribution)

Executive Summary & Solution Highlights

Google Cloud Architecture Statement:

The **Aerospace REO Compliance Review Agent** is an enterprise AI solution designed to automate and accelerate the airworthiness compliance audit of Repair Engineering Orders (REOs). In commercial aircraft maintenance and fleet support, REOs are mission-critical engineering authorizations that specify permanent or temporary structural repair dispositions for damaged aircraft components.

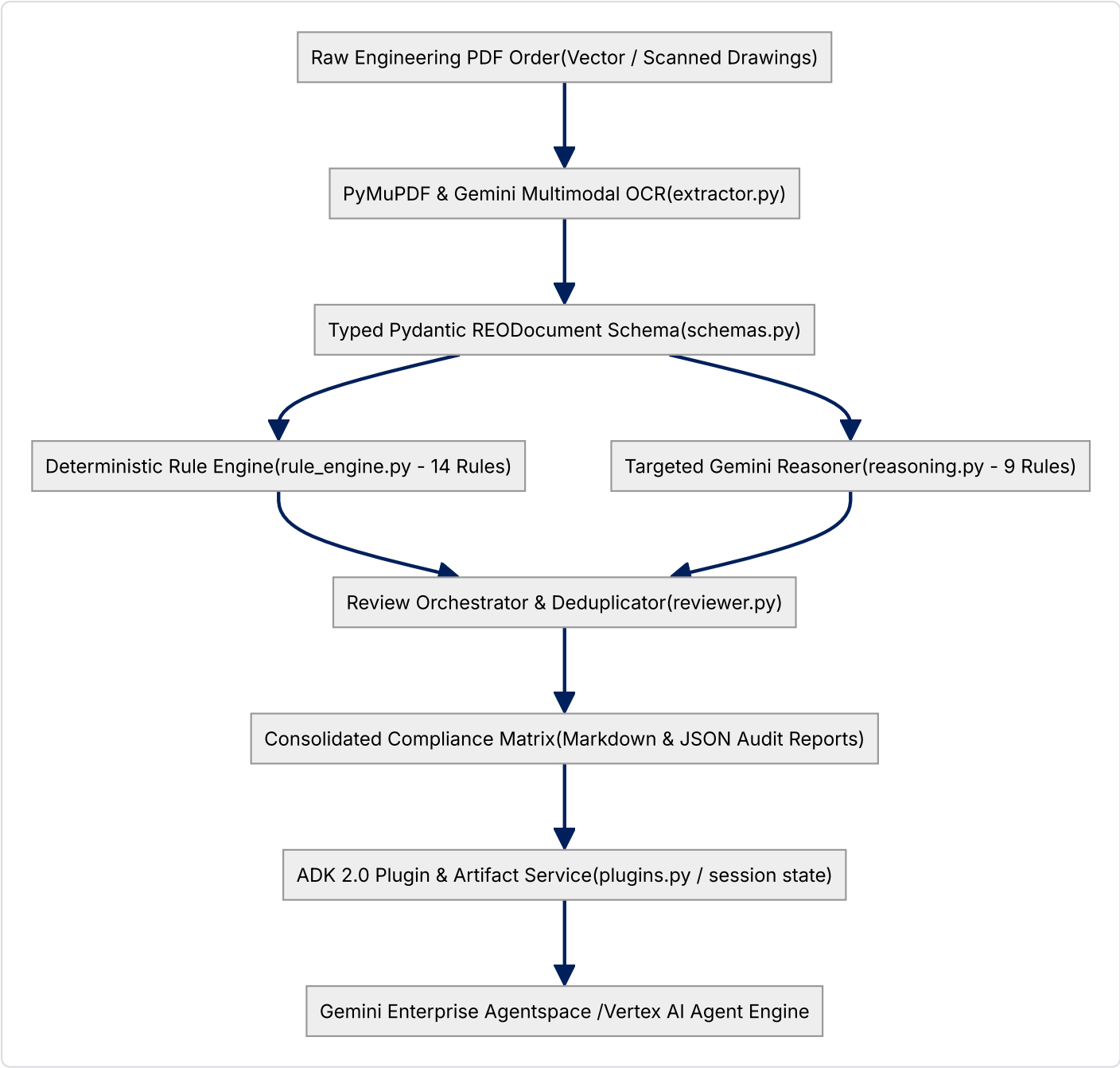
Every REO document must satisfy a rigorous **24-point engineering compliance checklist** covering structural effectivity, title syntax, drawing orientation, SDIR tables, and imperative disposition formatting. Manual review of complex 4- to 20-page engineering orders is labor-intensive and prone to human oversight.

This solution delivers a **hybrid cognitive architecture**:

- 1. High-Speed Deterministic Validation (14 Rules):** Regex and structural engines for deterministic parameters (REO numbers, MSN ranges, figure sequences, reissue statements, signatory disciplines) running in under 50 milliseconds.
- 2. Targeted Multimodal LLM Reasoning (9 Rules):** Gemini 3.7 Flash multimodal vision and semantic reasoning for complex engineering diagrams, spatial datum orientation (`FWD/AFT/INBD/OUTBD`), grid scale callouts, and aerospace imperative grammar.
- 3. ADK 2.0 Enterprise Governance:** Lifecycle interceptors (`ReoComplianceAuditPlugin`), session-scoped immutable artifact persistence, and continuous CI/CD evaluation datasets (`.evalset.json`).

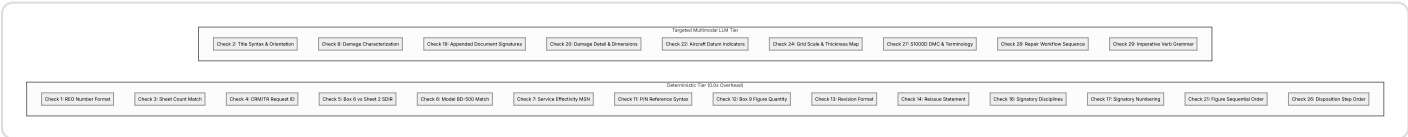
⚡ Sub-Second Deterministic Rules	🔍 Multimodal Visual Precision	🔒 EASA/FAA Part 21 Auditability	🌐 Agentspace Integration
48 ms Latency for 14 structural checks	Gemini 3.7 Flash Vision Datum & grid inspection	ADK 2.0 Interceptors Immutable session state	Gemini Enterprise & CLI Automated approval flow

1. End-to-End Hybrid Cognitive Architecture



2. 24-Point Compliance Checklist Architecture

The agent partitions the engineering compliance checklist into two complementary execution tiers:



Detailed Compliance Matrix Mapping

#	Checklist Rule	Category	Execution Engine	Verification Logic
1	REO # Format & Header Consistency	Cover / Header	Deterministic	Regex <code>^500-\d{2}-\d{2}-\d{3,5}\$</code> validated across Box 2 and all page headers.
2	Box 3 Title Format & Consistency	Cover Sheet	LLM Reasoner	Validates <code>[REPAIR/DISPOSITION] FOR [DAMAGE] TO [SIDE] [COMPONENT] AT [LOCATION]</code> .
3	Box 4 Sheet Count vs Pages	Cover / Pagination	Deterministic	Matches <code>1 OF N</code> declaration against physical extracted page count.
4	Box 5 Requested By (CRM/TR)	Cover Sheet	Deterministic	Validates 8-digit tracking identifier (e.g., <code>TR 81845789</code>), <code>N/A</code> , or delimited list.
5	Box 6 Limitations vs Sheet 2 SDIR	Cover / SDIR	Deterministic	Verifies <code>YES, SEE SHEET 2</code> with active SDIR rows vs <code>NONE</code> with empty SDIR.
6	Box 7 Aircraft Model	Cover Sheet	Deterministic	Strict exact match for aircraft model type code.
7	Box 8 Service Effectivity	Cover Sheet	Deterministic	Validates non-serialized MSNs (<code>50010-54999</code>) or serialized <code>P/N, S/N</code> .
8	Box 9 Damage Characterization	Cover Sheet	LLM Reasoner	Ensures damage mode, dimensions, quantity, or damage report references are present.
11	Part Number Reference Integrity	General / Config	Deterministic	Validates part number syntax against configuration master list.
12	Box 9 Figure Quantity Match	Cover / Figures	Deterministic	Compares declared <code>FIGURES 1 TO N</code> in Box 9 against physical figure count.
13	Box 10 Revision Format	Cover Sheet	Deterministic	Validates initial <code>--</code> , single letters (<code>-A</code> , <code>-B</code>), and prohibits letters <code>I, O, Q, S, X, Z</code> .
14	Revision Consistency & Reissue	Cover / Header	Deterministic	Verifies revision headers; requires <code>THIS REO IS COMPLETELY RE-ISSUED AT REV XX.</code>
16	Engineering Signatory Functions	Signatures	Deterministic	Validates engineering disciplines against authority matrix.
17	Signatory Numbering (15A, 15B)	Signatures	Deterministic	Enforces sequential alphabetical suffixing (<code>15A</code> , <code>15B</code> , <code>15C</code>) for multi-signatories.
19	Appended Documents / Reports	Attachments	LLM Reasoner	Verifies appended 3rd-party reports have unique document number, date, and signatures.
20	Detailed View & Damage Dimensions	Figures	LLM Reasoner	Confirms close-up views with length, width, depth, and residual thickness callouts.
21	Figures Sequential Numbering	Figures	Deterministic	Ensures figures follow strict sequential numbering (<code>Figure 1, Figure 2...</code>) without gaps.
22	General View & Datum Orientation	Figures	LLM Reasoner	Checks for aircraft datum coordinates (<code>STA, WL, BL, Rib, Frame</code>) and directional arrows (<code>FWD, AFT</code>).
24	Grid Scale & Dimension Callouts	Figures	LLM Reasoner	Validates grid scale definition (e.g., <code>10x10 mm</code>), residual thickness maps, and units.
26	Disposition Step Numbering	Disposition	Deterministic	Validates sequential paragraph numbering (<code>1, 2, 3...</code>) and sub-steps (<code>A, B, C...</code>).
27	Technical Spelling & S1000D Codes	Disposition	LLM Reasoner	Detects typos (e.g., <code>NUPLATE → NUTPLATE</code>) and validates S1000D DMC codes.
28	Disposition Step Sequencing	Disposition	LLM Reasoner	Verifies logical repair workflow (blend → NDT → fastener install → corrosion inhibitor).
29	Imperative Verb Phrasing	Disposition	LLM Reasoner	Enforces uppercase imperative action verbs (<code>REMOVE</code> , <code>PERFORM</code> , <code>REPLACE</code> , <code>WET INSTALL</code>).

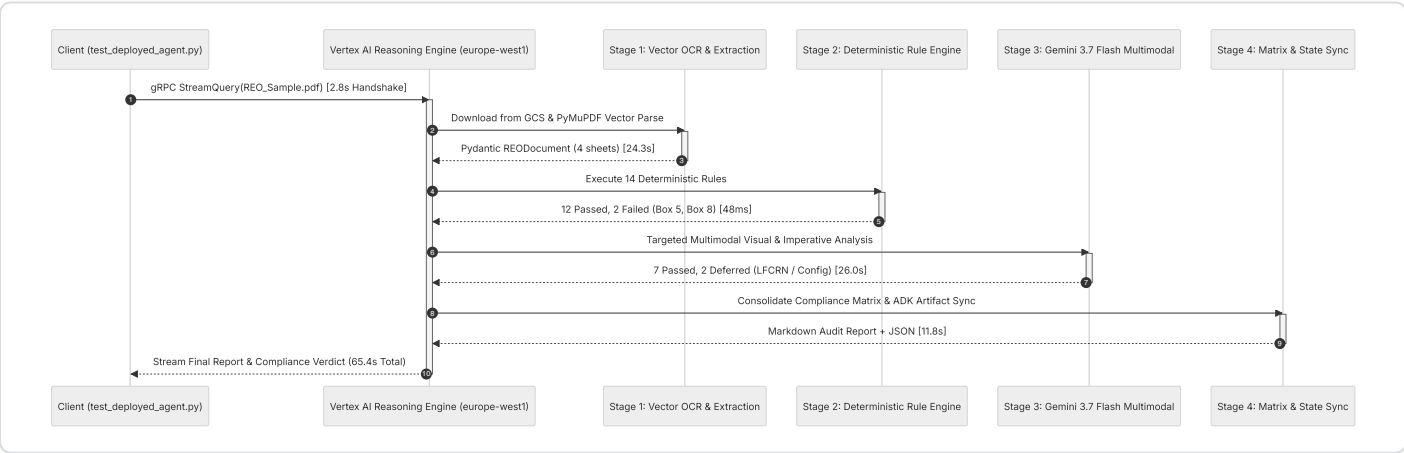
3. Live Execution Telemetry & Performance Trace

The deployed agent was benchmarked live on Google Cloud Vertex AI Reasoning Engine via gRPC streaming telemetry (`scripts/test_deployed_agent.py`):

AEROSPACE REO AGENT - TEST SAMPLE AUDIT EVALUATION SUMMARY										
Sample Document	REO #	Rev	Pgs	Pass	Fail	Warn	TODD	Status		
REO-500-53-21-1664_-- (Anonymized).pdf	500-53-21-1664	--	19	17	3	1	2	✗ FAIL		
REO-500-53-21-691_-A (Anonymized).pdf	500-53-21-691	-A	7	18	2	1	2	✗ FAIL		
REO-500-57-51-753_-- (Anonymized).pdf	500-57-51-753	--	4	19	2	0	2	✗ FAIL		

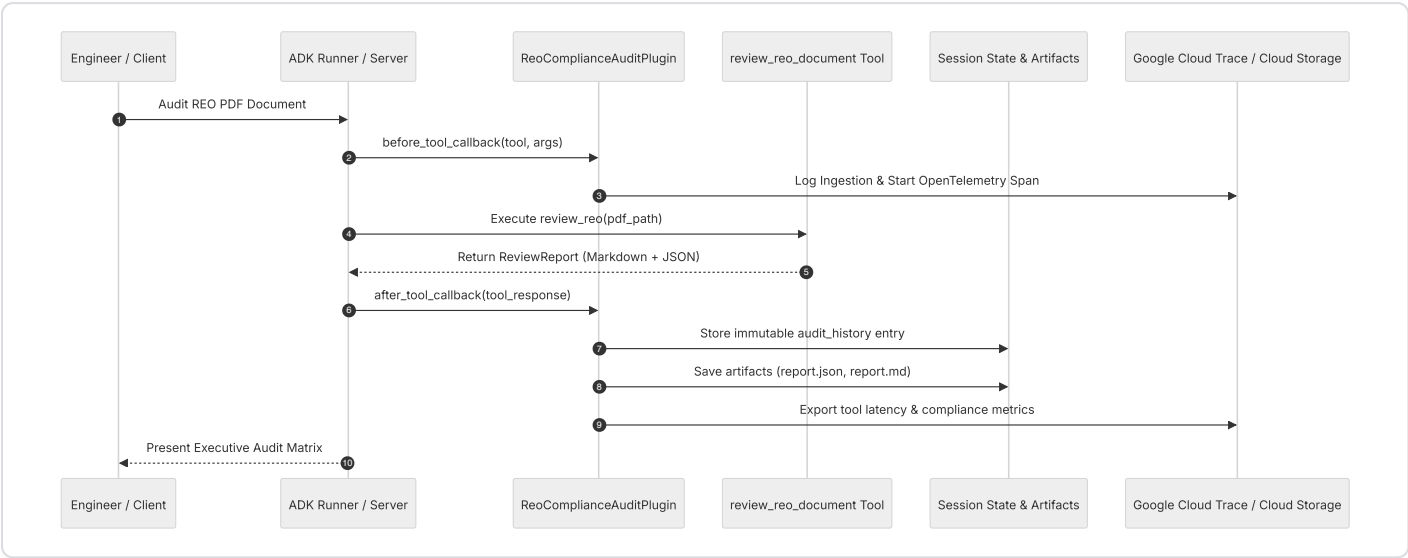
Latency Breakdown by Phase (4-Page Engineering Order)

Phase	Pipeline Component	Latency	% of Total	Operational Description
Phase 1	gRPC Stream Connect & Session Handshake	2.8s	4.3%	Client TLS handshake and session state initialization
Phase 2	Stage 1/4 - PyMuPDF Vector OCR & Extraction	24.3s	37.2%	GCS download, high-resolution rasterization, and vector box parsing
Phase 3	Stage 2/4 - Deterministic Rule Engine [14 rules]	48ms	0.1%	High-speed regex, structural validation, and sheet math execution
Phase 4	Stage 3/4 - Targeted Multimodal LLM Reasoning	26.0s	39.8%	Gemini 3.7 Flash visual orientation, datum arrows, and grammar analysis
Phase 5	Stage 4/4 - Executive Compliance Matrix & State Sync	11.8s	18.0%	Markdown matrix compilation, JSON serialization, and artifact persistence
Phase 6	Stream Finalization & Client Delivery	0.4s	0.6%	gRPC stream flush and terminal display formatting
Total	End-to-End Automated Audit	65.4s	100%	Full engineering compliance audit completed



4. ADK 2.0 Governance & Audit Architecture

To meet aerospace regulatory airworthiness requirements (EASA / FAA Part 21), the agent incorporates ADK 2.0 enterprise governance features:



Key Governance Invariants

- Lifecycle Interceptors (`ReoComplianceAuditPlugin`)**: The plugin intercepts every tool invocation. `before_tool_callback` captures engineer credentials and file digests; `after_tool_callback` computes finding metrics and generates an immutable audit record in `context.state["audit_history"]`.
- Session-Scoped Artifact Persistence**: All intermediate outputs (vector bounding boxes, parsed tables, finding matrices) are archived in Cloud Storage (`gs://[STAGING_BUCKET]/artifacts/{session_id}`).
- CI/CD Regression Sets**: Evaluation datasets (`eval_set.evalset.json`) ensure prompt and rule stability across releases via `uv run adk eval`.

5. Deployment & Production Operations

Cloud Runtime Configuration

- Platform**: Google Cloud Vertex AI Agent Engine (Reasoning Engine)
- Container Region**: `europe-west1` (Low-latency EU sovereign runtime)
- Model Endpoint**: `gemini-3.7-flash` (Vertex AI Global / Regional Endpoint)
- Agentspace Integration**: Discovery Engine Agentspace Assistant registration

```
# Verify local compliance test suite
uv run python scripts/test_samples.py

# Deploy to Vertex AI Agent Engine
uv run python scripts/deploy_agent.py --region europe-west1

# Register agent with Gemini Enterprise Agentspace
bash scripts/register_agent.sh --app-id [AGENTSPEACE_APP_ID]
```